





An Agentic Framework Using Slime Mold Intelligence for Transportation Network Optimization

FINAL REVIEW

TEAM MEMBERS

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PROJECT GUIDE

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BATCH NO: 09

INTRODUCTION

- Designing transport networks requires balancing infrastructure costs, user efficiency, and fault tolerance.
- Poor route planning in urban and global logistics networks leads to increased operational costs and longer travel times.
- The slime mold **Physarum polycephalum** naturally grows decentralized, highly efficient, and resilient network pathways.
- The proposed model mimics this organism's behavior to automatically strengthen optimal routes and eliminate weaker links.
- This project applies these bio-inspired algorithms to optimize Indian transport corridors and broad supply chain logistics.

ABSTRACT

Urban transportation and global logistics networks require massive investments, where inefficient route planning inevitably leads to higher operational costs and longer travel times. To address this challenge, our proposed project simulates the behavior of the slime mold Physarum polycephalum, an organism naturally adept at constructing highly efficient and resilient pathways. By deploying bio-inspired computational algorithms, the system evaluates transport hubs as discrete nodes, automatically strengthening optimal routes while pruning weaker paths. Ultimately, this nature-inspired approach aims to evaluate large-scale Indian transit corridors and supply chains, providing planners with reliable, sustainable methods to design smarter, future-ready networks.

LITERATURE SURVEY

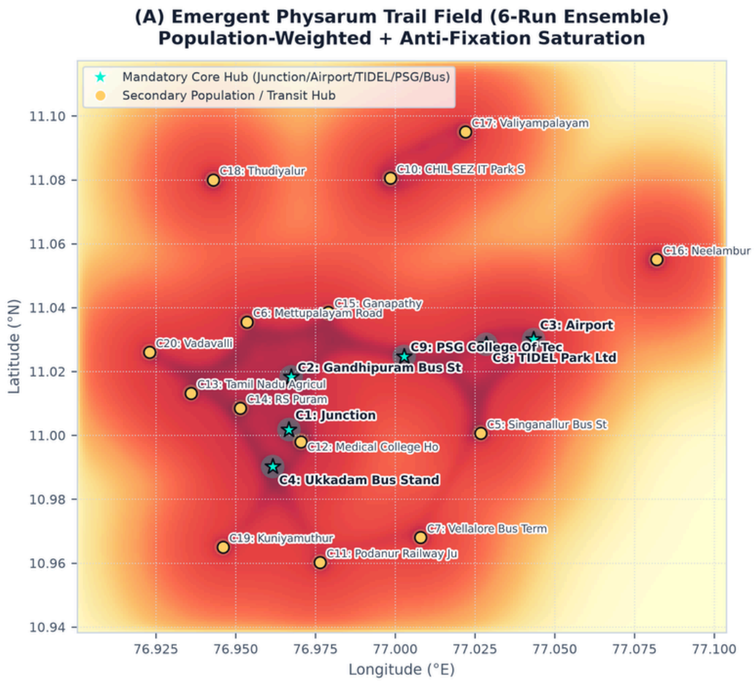
S. No	Authors & Year	Title	Methodology	Limitation	Your Contribution
1	Chu et al. (2021)	A Physarum-Inspired Algorithm for Logistics Optimization: From the Perspective of Effective Distance	Effective-distance-based Physarum logistics optimization	Limited consideration of real-world geographical constraints	Generate spatial metro links using Jones agents.
2	Wang et al. (2022)	Design a Robust Logistics Network with an Artificial Physarum Swarm Algorithm	Artificial Physarum swarm for robust logistics networks.	Computational complexity may limit scalability to large networks	Assess link stability across repeated runs.
3	Awad et al. (2023)	A Novel Physarum-Inspired Competition Algorithm for Discrete Multi-Objective Optimisation Problems	Physarum-inspired multi-objective optimization.	Limited focus on spatially explicit transport network generation	Evaluate spatial network generation across runs.
4	Jones, 2010	Characteristics of Pattern Formation and Evolution in Approximations of Physarum Transport Networks	Multi-agent particle system on rasterized 2D maps.	High computational burden for convergence	Adopt as the core engine, dynamically optimizing sensory parameters for faster convergence on Indian transit maps.
5	Heeroo & Gukhool, 2024	PANDA: A Physarum-Inspired Algorithm to Solve the Multi-Objective Discrete Network Design Problem	PANDA: multi-objective Physarum network design.	Computational complexity may hinder large-scale network design	Apply Jones-style agents to our problem.

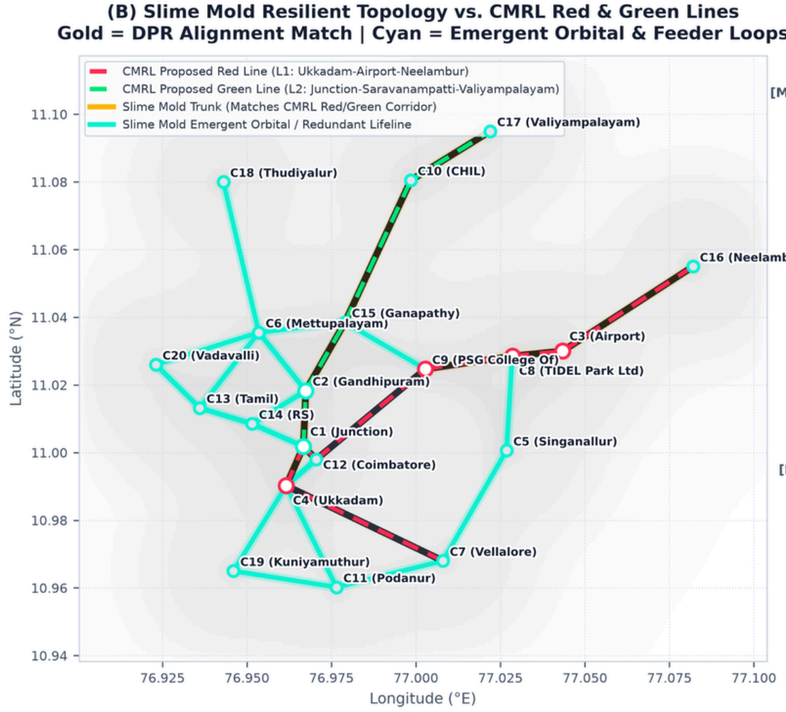
WHY NOT USE GOOGLE MAPS??

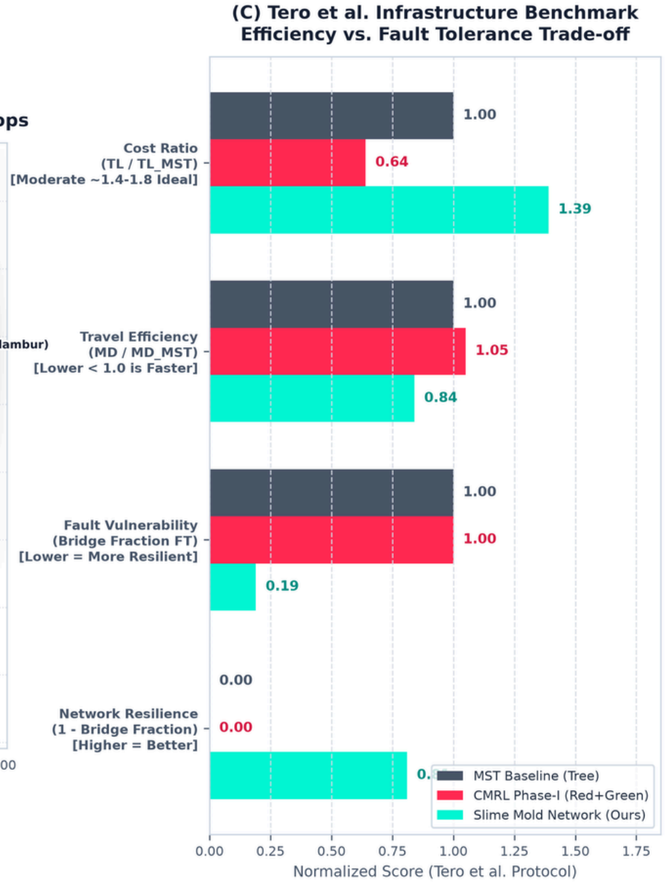
1. **Beyond Existing Routes:** Google Maps and standard shortest-path algorithms primarily find routes through existing networks; our goal is to explore alternative connections for a new, greenfield metro network.
2. **Population-Aware Connectivity:** Instead of optimizing distance alone, our approach incorporates population-weighted hubs to prioritize connectivity to important urban areas.
3. **Multi-Objective Optimization:** Infrastructure planning involves trade-offs between total network length, travel efficiency and fault tolerance—not just finding the shortest path between two points.
4. **Emergent Network Design:** Unlike a simple predefined route or fixed topology, the Physarum-inspired algorithm adaptively reinforces useful connections and weakens less useful ones to explore alternative network structures.
5. **Resilience Beyond Efficiency:** A network may be efficient under normal conditions but vulnerable when a critical connection fails. Our approach evaluates network resilience alongside efficiency and infrastructure cost.

THE COIMBATORE EXAMPLE

COIMBATORE GREENFIELD METRO DESIGN: BIO-INSPIRED PHYSARUM POLYCEPHALUM VS. CMRL PHASE-I DPR







RESEARCH GAP & PROJECT POSITIONING

What Prior Work Often Assumes:

- Evaluates models mostly on predefined graphs or static, pre-existing networks in developed regions.
- Treats geographical terrain as a uniform plane, ignoring dynamic real-world physical obstacles.
- Emphasizes theoretical biological mimicry rather than practical, real-world infrastructure deployment.
- Fails to address urgent, dynamic logistics such as disaster evacuation routing.
- Overlooks the specific challenge of designing resilient links to geographically isolated communities.

What This Project Addresses:

- Integrates real-world spatial constraints, topographical barriers, and variable-density hubs directly into the agent environment.
- Simulates adaptive emergency evacuation corridors that actively self-repair and divert around spreading hazards.
- Discovers optimal, cost-effective lifeline routes to connect isolated and remote settlements.
- Fine-tunes agent sensory parameters to balance construction length, transit efficiency, and fault tolerance.
- Delivers a fast, nature-inspired decision-support tool for emergency management and resilient regional transit planning.

SYSTEM ARCHITECTURE

Geospatial Dataset

Collects latitude and longitude coordinates of target transport hubs.

Gird Mapping

Converts the real-world geographic map into a two-dimensional computational grid.

Simulation Initialization Module

Plots the target transport hubs on the grid as discrete "nutrient sources" (food)., and inoculates the starting central node

The Slime Mold Algorithm

Attractant Diffusion
Agent Foraging & Growth
Flux-Conductance Feedback

Network Extraction & Optimization

Extracts the final stabilized tubular network connecting all designated hubs.

Evaluation & Benchmarking Module

Compares the simulated slime mold network against the actual as-built infrastructure.

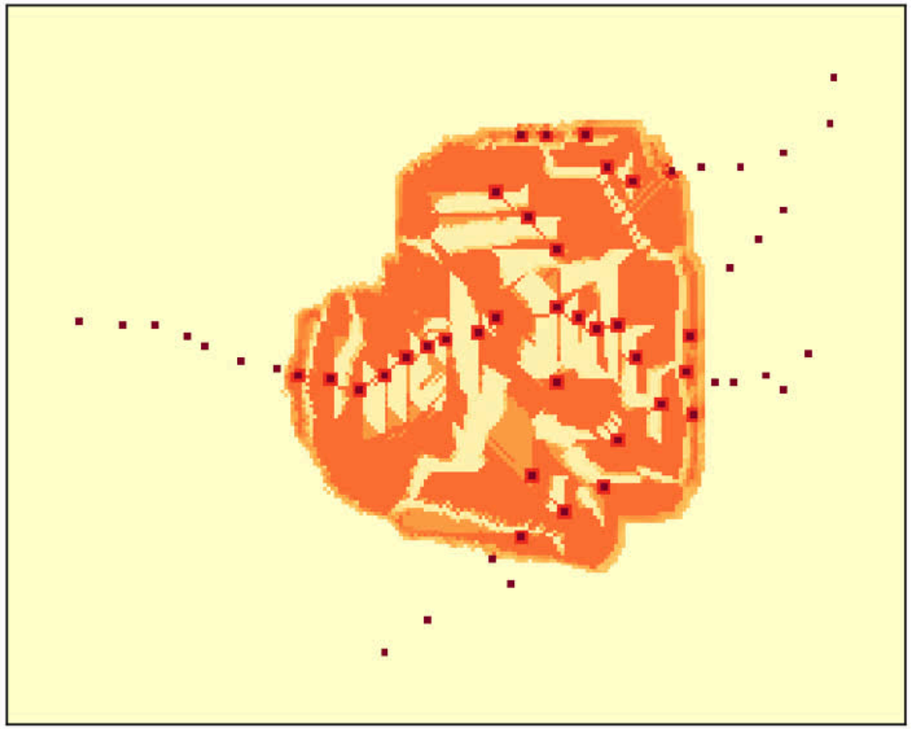
SOFTWARE SPECIFICATION

- **Environment & OS:** Windows 11 / Debian 13, utilizing VS Code as the primary IDE.
- **Core Languages:** Python 3.10+ (for rapid prototyping and data extraction) transitioning to Rust (for production-grade, memory-safe, multi-agent parallelism).
- **Geospatial & Map Parsing:** OSMnx and GeoPandas (Python) / osmpbfreader and geo (Rust) for extracting, cleaning, and translating real-world terrain and transit hubs into computational grids.
- **Simulation Engine:** rayon (Rust) for thread-safe CPU parallelism and ndarray (Rust) / NumPy (Python) for efficient multi-dimensional array processing of chemoattractant diffusion.
- **Visualization & Metrics:** macroquad (Rust) for fast 2D real-time rendering of agent populations, combined with Matplotlib and Seaborn (Python) to plot evaluation metrics (Total Length, Shortest-Path, Fault Tolerance).

HARDWARE SPECIFICATION

- **Processor (CPU)**: Multi-core processor (Intel Core i7 / AMD Ryzen 7 or higher) for handling concurrent execution of thousands of simulated agents and diffusion equations.
- **System Memory (RAM)**: 16 GB DDR4/DDR5 (32 GB recommended) to prevent memory overflow when staging large-scale, high-resolution rasterized geographic maps and agent matrices.
- **Storage**: 512 GB NVMe M.2 SSD for fast input/output (I/O) throughput when loading raw OpenStreetMap (OSM) datasets and exporting simulation frames.
- **GPU Acceleration:** Entry-level GPU (NVIDIA RTX 3060 or equivalent) to support potential compute shader offloading for heavy 2D grid rendering.

SIMULATION DEMONSTRATION

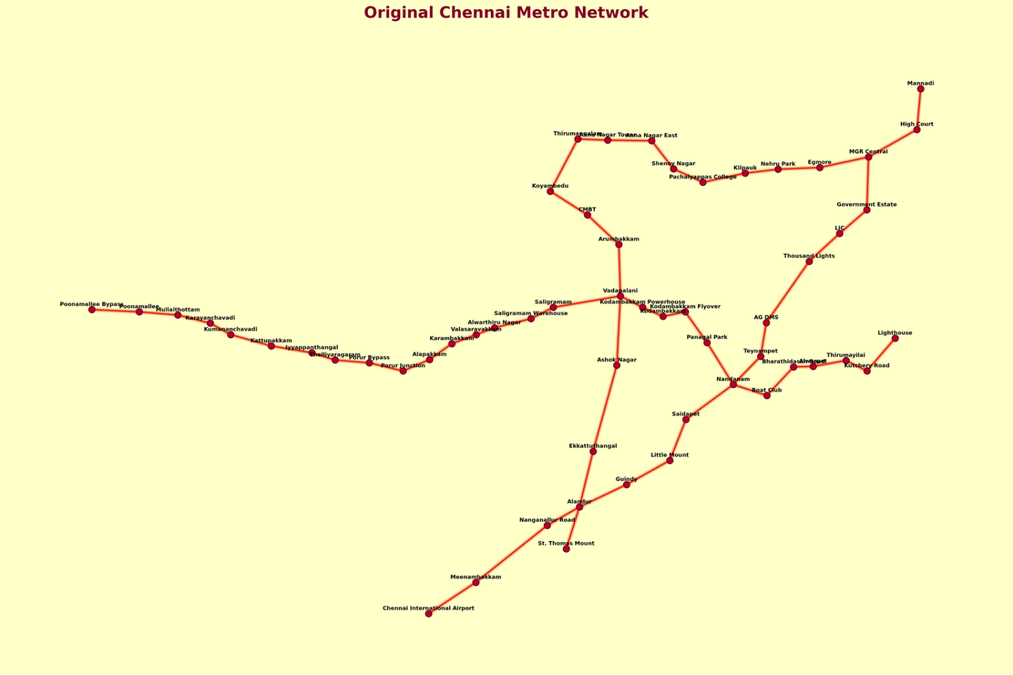


Simulation in Progress of simple Slime Mold algorithm

SIMULATION RESULTS

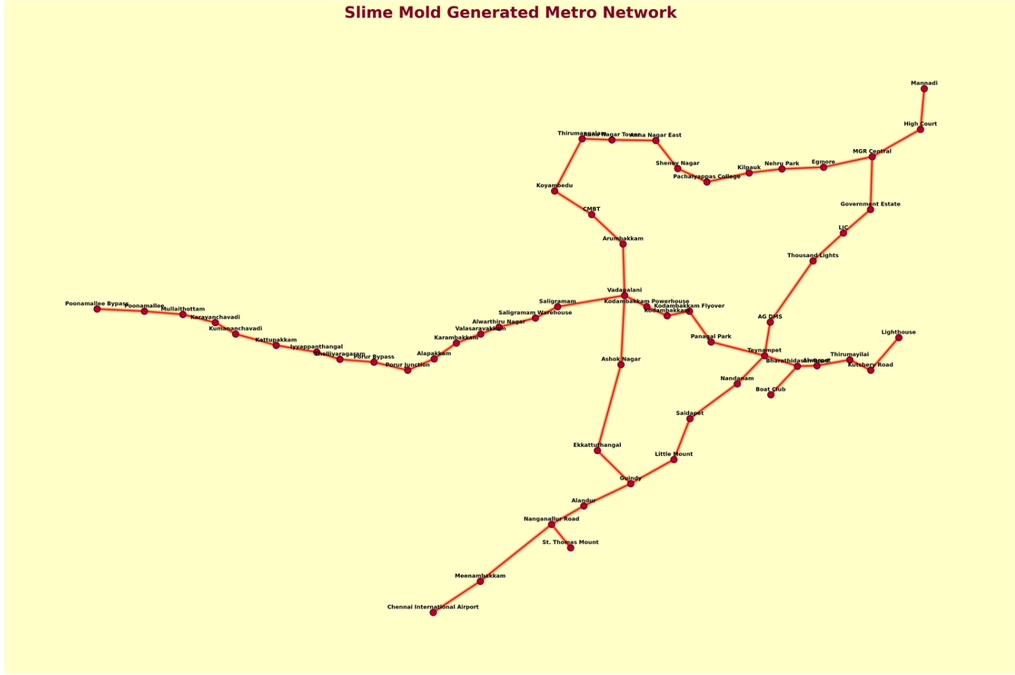
CURRENT NETWORK

Existing Chennai metro network

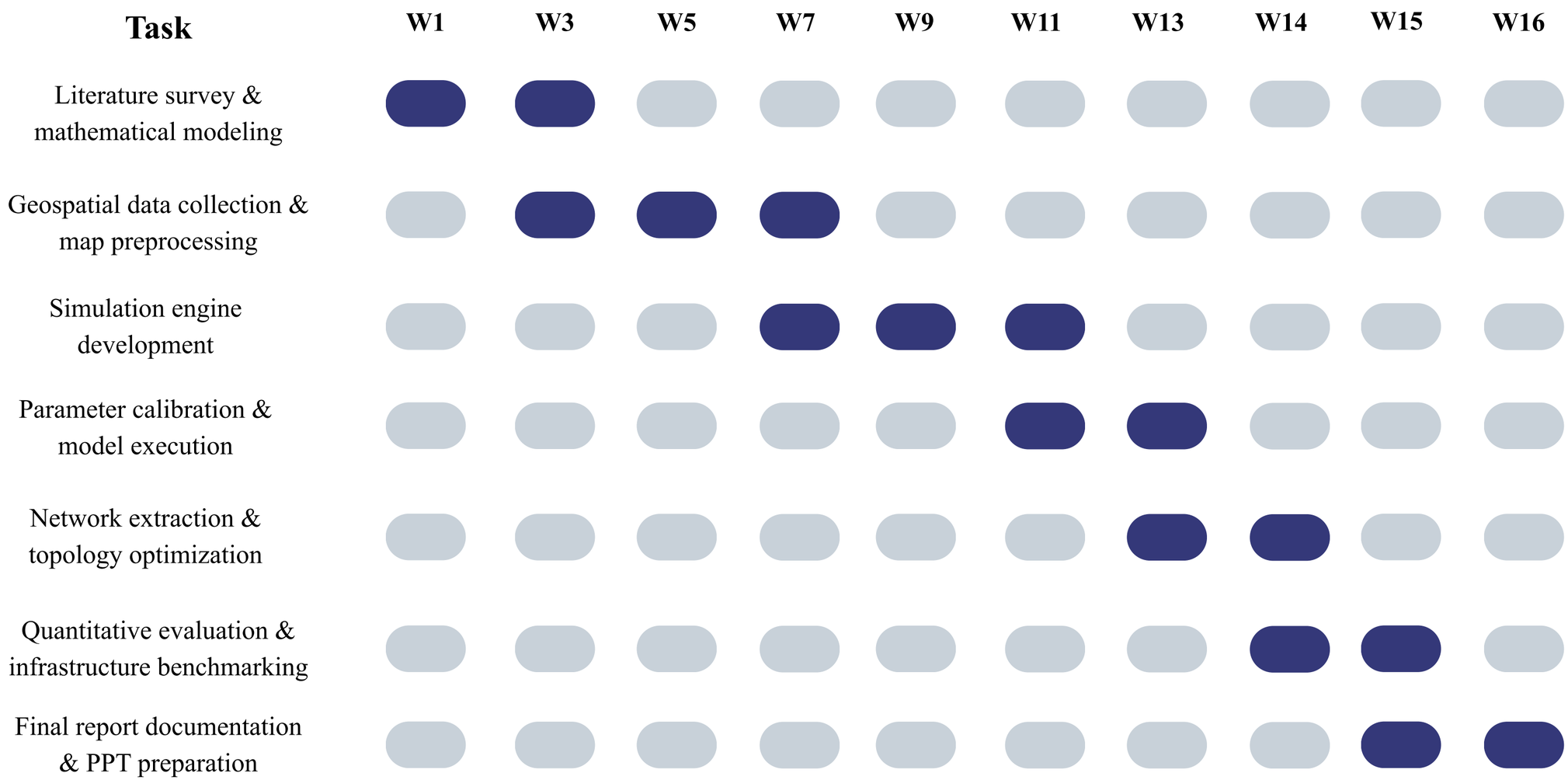


GENERATED NETWORK

Network Generated by Simulation



PROJECT GANTT CHART



CONCLUSION

"Our model bridges biological intelligence and urban planning—turning the natural foraging behavior of slime mold into actionable, resilient routing strategies. By balancing infrastructure cost, transport efficiency, and fault tolerance, it provides a proactive tool for sustainable transit design and emergency management."

Future Works:

- Migrate the core simulation engine to Rust for massive-scale, high-performance parallel execution.
- Integrate real-time dynamic datasets (e.g., live traffic updates, expanding disaster zones) into the agent environment.
- Develop an interactive, user-friendly graphical interface (GUI) tailored for city planners and emergency responders.
- Expand the simulation to account for 3D topographical elevations and multi-modal transport networks.
- Deploy the model on targeted real-world pilot case studies, such as the Chennai Metro expansion or coastal disaster evacuation routes.

THANK YOU